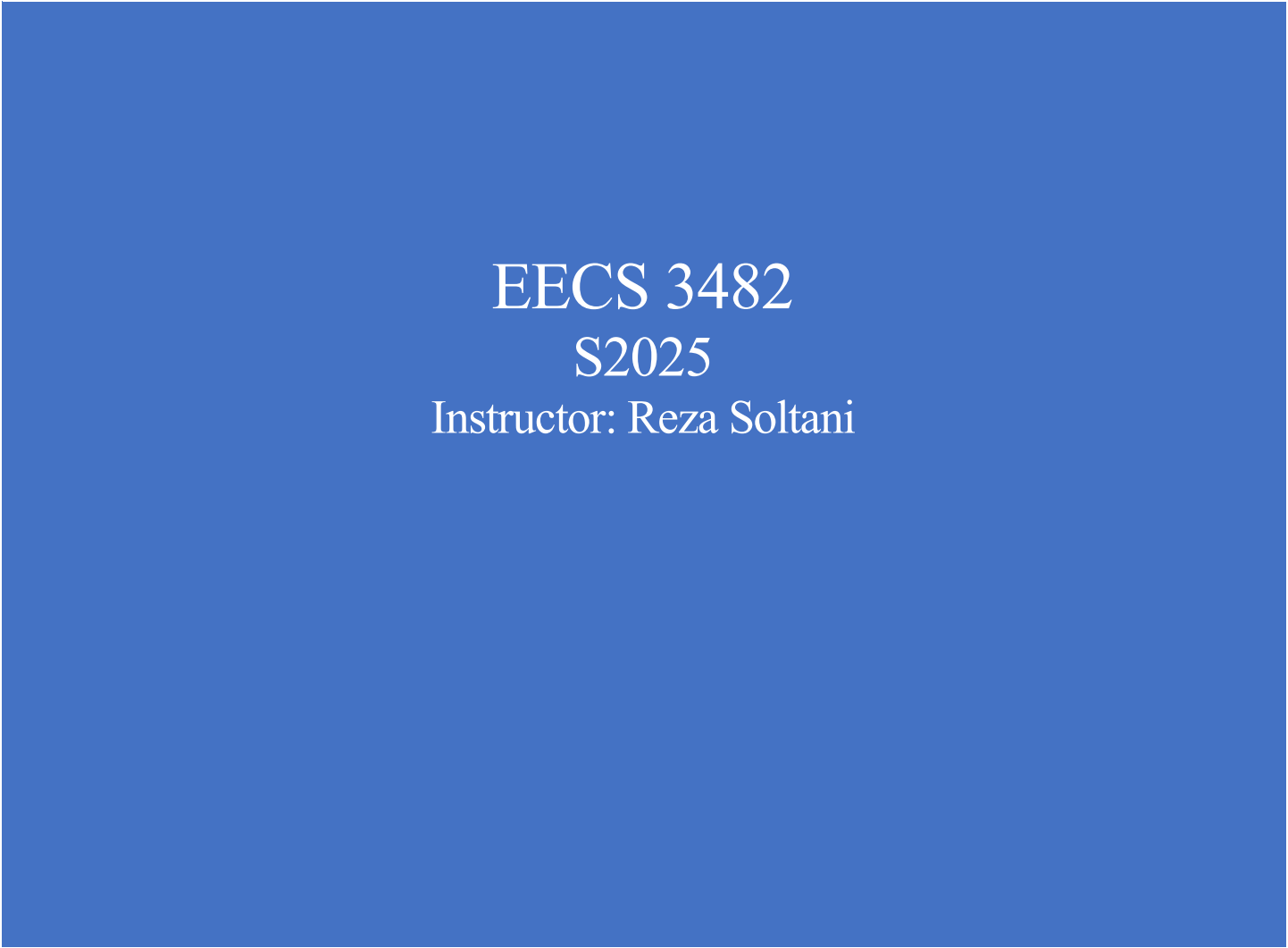




Lab 5 – File and Network Analysis



EECS 3482
S2025
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Lab Format

For every step of your labs, including this lab and all future labs in the course, you must provide a clear and **legible screenshot** whenever you use the command prompt.

Before taking the screenshot, and either before or after completing the step, you **must** enter the following command so that your **username**, **full name**, and **timestamp** are visible in the screenshot:

```
echo " Step completed by $(whoami), name: <first> <last>, on $(date)"
```

Note, replace *<first>* and *<last>* with your own first name and last name before running the command.

For example, a student with the name *John Smith* and username *JSUser1* should run the following command:

```
echo "Step completed by $(whoami), name: john smith, on $(date)"
```

to get the following result:

```
Step completed by JSUser1, name: John Smith, on Sat 17 May 2025 12:48:05 EDT
```

Note, there are different kinds of shell interface within Unix/Linux systems. Depending on the environment you are in (such as the *eeecsvm*'s kali environment which uses **ksh** instead of **bash** shell), you may have to enter the following modified command:

```
echo " Step completed by `whoami`, name: <first> <last>, on `date`"
```

or alternatively, enter **bash** in the command prompt to switch to *bash* shell before entering the original *echo* command.

Also note: If you are working on a Windows machine, you must use an equivalent command such as:

```
echo Step completed by %USERNAME%, name: <first> <last>, on %DATE% %TIME%
```

If a lab step does not include the above command or a clear screenshot, marks will be deducted.



Section 1 - YARA

In the first section, we're going to get familiar with [YARA](#)!

From YARA's website: [YARA](#) is a tool aimed at (but not limited to) helping malware researchers to identify and classify malware samples.

Important note: make sure not to execute the malwares you download in this section.

Part 1: Setting Up the Malware Analysis Environment

Step 1: Create an Isolated Malware Lab

Since we will be working with real malware samples, it is essential to create an **isolated and controlled environment**:

1. Open a **terminal** in **Kali Linux** *ideally* outside of your own machine.
2. Create a directory for malware analysis, call it: **malware_lab**

If not available download YARA on Kali using:

```
sudo apt update  
sudo apt install yara -y
```

Part 2: Downloading a Malware Sample

We will use freely available malware from reputable research sources.

Step 2: Obtain a Malware Sample

You can download real malware samples from these sources:

1. **MalwareBazaar** (<https://bazaar.abuse.ch>)
 - A collection of malware samples categorized by type.
 - Search for common malware (e.g., RedLine Stealer, AgentTesla).
 - Download a **password-protected ZIP file (password: infected)**.
2. **theZoo** (<https://github.com/ytisf/theZoo>)
 - A repository containing real-world malware for research.
 - Clone the repository, or download directory

Step 3: Extract the Malware Sample

1. Move the downloaded malware sample to your **malware_lab** directory.
2. Extract the files

Warning: The malware file (e.g., malware.exe) is now extracted.

Do NOT execute the malware files!

Part 3: Static Malware Analysis with YARA

Step 4: Inspect the Malware File

In this step, we're going to use various tools to learn more about the characteristics of the malware that you've downloaded. Check for the following 7 type of information and document your results.

Depending on the type of malware you have downloaded you may get all or some of the information below.

Programs/commands such as **file**, **strings**, **objdump**, **exiftool**, **binwalk**, **rabin2**, **readelf** and **pecheck** may be useful for the following 6 types of information!

1. Identify the file name, type and format (ex. Window PE, ELF, script, ...)
2. Extract readable strings from the binary:
 - Look for **suspicious and unique keywords**, such as:
 - Hardcoded URLs (e.g., <http://malicious-site.com>)
 - Registry modifications (e.g., HKEY_LOCAL_MACHINE)
 - Process injections (VirtualAlloc, CreateRemoteThread)
3. Check for unique cryptographic functions:
 - Embedded encryption keys, base64-encoded strings, or obfuscation techniques.
4. Malware's imported functions:
 - Which Windows API functions the malware relies on (e.g., CreateProcessA, WriteFile).

5. Identify and extra metadata from the file:
 - metadata such as timestamps, compiler information, and version details.
6. Check for embedded payloads and hidden files.

Step 5: Create a YARA Rule

We will now create a **custom YARA rule** to detect your malware.

1. Create a new YARA rules file in **an editor such as Nano**. Call the new file **my_malware.yara**.
2. Add the following YARA rule structure as the starting point:

```
rule MyMalware
{
    meta:
        description = "Detects a sample malware file"
        author = "Your Name"
        date = "2025-03-19"

    strings:
        $string1 = "malicious.com" nocase
        $string2 = "VirtualAllocs"
        $string3 = { E8 83 C4 04 89 45 AB }

    condition:
        any of them
}
```

3. Explanation of the rule:
 1. **Metadata Section:** Describes the rule (author, description, date).
 2. **Strings Section:** Defines patterns to search for in malware.
 1. \$string1: Detects a malicious domain name.
 2. \$string2: Identifies a Windows API call often used in malware.
 3. \$string3: Looks for a specific hex pattern within the malware binary.
 3. **Condition:** The rule triggers if any of these patterns are found.

Refer to <https://yara.readthedocs.io/en/latest/> for more details about YARA Rule files

4. Update the YARA rule file with strings and the condition that matches the strings you found in Step 4.

Note the sample YARA file above is just an example. Your YARA file should look different. The strings in your YARA rule file should be obtained from your experiments in Step 4. Try your best to pick strings that can uniquely identify your malware.

5. Save your rule file and exit.

Step 6: Run YARA Against the Malware Sample and your YARA rule file

Explore YARA's manual to find the command line arguments necessary to complete the following 2 steps:

1. Check your YARA rule file's syntax to make sure it's valid! There is a YARA command line switch allows you to do just that. YARA will not work properly if you have a typo or a mistake in your YARA rule file.
2. **Using the YARA program, check the malware file against the your YARA file.** Include the necessary YARA flag to show which strings in the malware matched the rules in your YARA file!
3. Finally, create SHA256 of your malware by entering:

sha256sum [malware file]

Make sure to include every step of the lab in your report. Include the link to the malware, the malware's details, the information you were able to obtain about the malware in Part 3-Step 4, the custom YARA rule file you created in Part 3-Step 5 and finally the result of checking your YARA file against the malware (Part 3-Step 6). The goal is to create a YARA file with rules that are as unique as possible that is able to successfully recognize your malware.



Section 2 - Snort

In this section we will discuss Intrusion Detection Systems using *Snort*. Snort is a free and open source network [intrusion detection system](#) (IDS) and [intrusion prevention system](#) (IPS). To learn more about Snort see:

<https://www.snort.org/>

Note: Some of the instructions may be different if using a more recent version of Snort (3.0+)

Check this link for more info about Snort installation and setup:

<https://gist.github.com/lukeswitz/08ea69ad6047c5f0bd2388476b2fd189#snort-3-installation-and-configuration-guide-for-debian-based-systems> and <https://www.howtoforge.com/install-and-configure-snort-3-on-ubuntu-22-04/amp/>

Goal

In this section, we will setup Snort on a Linux server to identify and alert certain types of network traffic. This part will show the foundation on which more comprehensive IDS and IPS configurations can be designed.

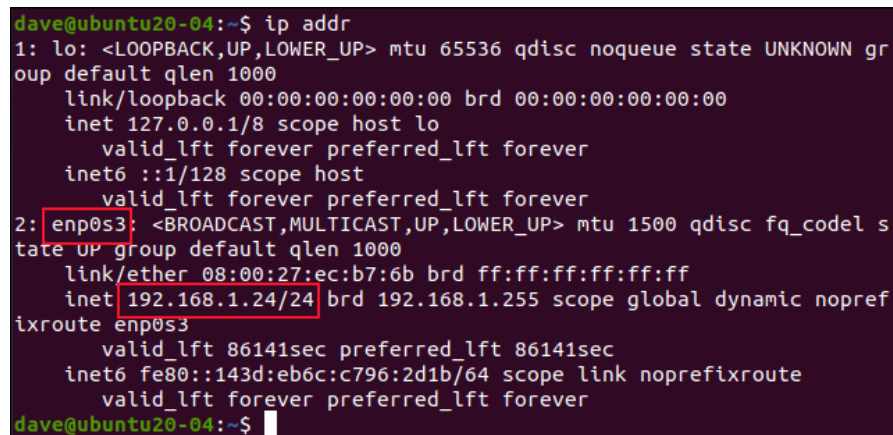
Part 1 – Install and Run Snort on your machine

- 1- Install Snort on a Linux machine. You can use the *Azure Cloud* lab environment to install Snort on Kali VM itself. Alternatively, you can install Snort on your local Linux environment to complete this lab. Note you can also install Snort on Windows, although the steps in this lab only cover the Linux version.

- a. To install Snort enter the following commands in the Linux terminal:
 - i. `sudo apt-get update`
 - ii. `sudo apt-get install snort`
- b. During the installation you may be asked a couple of questions about your network parameters. You must provide **your network interface name (ex. `ensp0s3`)** and **network address range or CIDR notation (ex. `192.168.1.0/24`)** to the installer. You can obtain these values by entering this command in the Terminal:

- i. `ip addr`

The result should look like:



```
dave@ubuntu20-04:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ensp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ec:b7:6b brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.24/24 brd 192.168.1.255 scope global dynamic noprefixroute
        valid_lft 86141sec preferred_lft 86141sec
    inet6 fe80::143d:eb6c:c796:2d1b/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
dave@ubuntu20-04:~$
```

Note, the above commands do not work in every Linux environment. If you cannot use 'apt-get' to install Snort, then follow the links below to install Snort using other methods, including through the source code compilation.

- Link 1: <https://linuxide.com/install-snort-on-ubuntu/>
- Link 2: <https://bin3xish477.medium.com/installing-snort-on-kali-linux-9c96f3ab2910>
- Link 3: <https://www.cloudsavvyit.com/6424/how-to-use-the-snort-intrusion-detection-system-on-linux/>
- Link 4: <https://www.youtube.com/watch?v=ALeQUhpUnqQ>

You can also Google *Snort + [Your OS Version]* to learn more about how to install Snort in your environment.

- c. Test successful installation of Snort by entering the command:
 - i. `snort -version`

- 2- Verify (and optionally configure) Snort setting by opening the 'snort.conf' file. This file is generally located at '/etc/snort/snort.conf'
 - a. **Note: in more recent version of Snort (3.x) the config file is a LUA file located at '/usr/local/etc/snort/snort.lua'**
 - b. Once the file is open, search for the line starting with: 'ipvar HOME_NET'. This line indicates the network address that you are protecting using Snort. Verify that the correct network address range is provided. Make necessary change if required. Then, save your changes and close.
 - c. **Note: The location of the config and rule files depends on the version of the operating system! Check the documentation for your specific OS.**
- 3- Snort is a signature-based IDS, there it allows you to defines custom rules to detect intrusion activities. Run the following command to get a list of existing rules:
 - a. `ls /etc/snort/rules`
Note: For Snort 3.x uses a different structure for rules and rule files are referenced in a Snort LUA file.
- 4- It is recommended that you update your Snort rules from Snort's download page here: <https://www.snort.org/downloads> , under the Rules section.
 - a. Download the rule set for the version of the Snort you have installed. If you do not have a Snort account (i.e. you are not registered), download the Community version of the rules.
 - b. Once the download is complete, enter the following command to extract the rules and install them into the right folder:
 - i. `sudo tar -xvzf snort3-community-rules.tar.gz -C /etc/snort/rules`
 - ii. **Update the above file name and path as required**
- 5- Start Snort by entering the command:
 - a. `sudo snort -d -l /var/log/snort/ -h [IP ADDRESS RANGE] -A console -c /etc/snort/snort.conf`
 - i. Replace **[IP ADDRESS RANGE]** with your network IP range you are protecting (ex. 192.168.1.0/24)
 - ii. **-A console:** allows the alerts to go to the console
 - iii. **-c /etc/snort/snort.conf:** defines which configuration should be read
 - iv. **-l /var/log/snort/:** specifies where the logs are stored
 - v. **-d:** Filters the application layer packets

Note: For Snort 3.x you should use a different command similar to:

- a. `sudo snort -c /usr/local/etc/snort/snort.lua -R /path/to/local.rules -A alert_fast`

You may need to include the -i argument to specify the network interface. Check Snort documentation for more details

Describe the results you get based on the steps completed so far, including any errors. Describe the parameters passed to Snort and what they do. Show a screenshot of the output you get from Snort.

Part 2 – Add a custom ICMP Rule

- 1- We are going to write a custom rule by updating the rules file.

Note: The location of the rules files may have changed in Snort 3.x. Check /usr/local/etc/snort/rules/local.rules or /etc/snort3/rules/local.rules if using Snort 3.x.

Otherwise, open the file '/etc/snort/rules/local.rules' and add the following to the file:

- a. alert icmp any any -> any any (msg:"ICMP Packet found"; sid:1000001; rev:1;)

The structure of a rule is:

BASIC OUTLINE OF A SNORT RULE

```
[action][protocol][sourceIP][sourceport] -> [destIP][destport] ( [Rule options] )
```

Rule Header

See links below for more detail about Snort rules.

The above rule defines that an **alert** will be logged if an **ICMP** packet is found. The ICMP packet could be from **any** IP address and the rule ID is **1000001**. Make sure to pick a **SID greater 1000000** for your own rules.

- 2- Start Snort using the same command as step 1, but this time specify the custom rule as input. Optionally remove the -d parameter.
- 3- Find your VM's IP address by typing 'ifconfig' and attempt to send an ICMP packet by entering:
 - a. ping [target_ip]
[target_ip] should be **in** the network range that Snort is monitoring. The easiest test is to ping the same server as the one Snort is running on.
- 4- After you send the ping message, an alert should be triggered. You can find the log messages in '/var/log/snort/snort.log'. For Snort 3.x the log file is changed to /usr/local/var/log/snort/.

Note: To stop Snort from scanning, by hitting CTRL + C, you may need to wait a few seconds to minutes for the log file to be created.

Note: the snort.log file may be in binary format. If so, you need to use **u2spewfoo** to read it. Simply pass the path of the log file to that program to translate and read its content.

Note: Snort may not be able to capture the ICMP packet sent and received from the same machine. To make ICMP work, you have to ping the VM from the other VM. I.e. if you try to ping from the same VM (i.e. ping itself) it may not work.

Describe the commands you executed and the results you obtained, including any errors.

Part 3 – Add a custom Rule for Google.com

- 1- Write a rule that will trigger when you browse the site: **google.com** from the machine Snort is monitoring.

Describe the commands you executed and the results you get, including any errors.

Troubleshooting Notes

If the error you see is about the network card or about the IP address not being reachable, then you can narrow down your search but providing a more specific IP address to your rules.

The network interfaces usually ignore traffic that isn't destined for their IP address. you can set your network card to *promiscuous* mode so that you can listen to all traffic within network range. The following command :

```
sudo ip link set enp0s3 promisc on
```

sets the network card with ID *enp0s3* to promiscuous mode.

If you are running Snort in a virtual machine, also remember to adjust the settings in your host for the virtual network card used by your virtual machine. For example, in VirtualBox,

you need to go to Settings > Network > Advanced and change the “Promiscuous Mode” drop-down to “Allow All.”

Snort References:

- <https://www.howtoforge.com/install-and-configure-snort-3-on-ubuntu-22-04/amp/>
- <https://gist.github.com/lukswitz/08ea69ad6047c5f0bd2388476b2fd189#snort-3-installation-and-configuration-guide-for-debian-based-systems>
- <https://cdn.comparitech.com/wp-content/uploads/2019/07/Snort-Cheat-Sheet.pdf>
- <https://snort.org/resources#documents> [Intro to Rule Writing]
- https://paginas.fe.up.pt/~mgi98020/pgr/writing_snort_rules.htm
- <https://dev.to/sankethj/detect-dos-ping-etc-using-snort-4gab>

Information about Snort 2.x

- <https://www.snort.org/snort2>
- <https://youtu.be/U6xMp-MIEfA?feature=shared>
- <https://www.youtube.com/watch?v=zZdYgglq8wc>